

```
mysql> create database practice2;  
Query OK, 1 row affected (0.01 sec)
```

```
mysql> show databases;  
+-----+  
| Database |  
+-----+  
| bookflow_db |  
| information_schema |  
| mysql |  
| performance_schema |  
| practice1 |  
| practice2 |  
| studentdb1 |  
| sys |  
+-----+  
8 rows in set (0.00 sec)
```

```
mysql> create table Employees (
-> EmpID int primary key,
-> EmpName varchar(50),
-> DeptID int,
-> Salary decimal(10,2),
-> Email varchar(100)
-> );
Query OK, 0 rows affected (0.03 sec)

mysql> insert into Employees (EmpID, EmpName, DeptID, Salary, Email) values (101, 'Rahul Sharma', 1, 55000, 'rahul.sharma@example.com');
Query OK, 1 row affected (0.01 sec)

mysql> insert into Employees (EmpID, EmpName, DeptID, Salary, Email) values (102, 'Priya Verma', 2, 45000, 'priya.verma@example.com');
Query OK, 1 row affected (0.01 sec)

mysql> select * from Employees;
```

EmpID	EmpName	DeptID	Salary	Email
101	Rahul Sharma	1	55000.00	rahul.sharma@example.com
102	Priya Verma	2	45000.00	priya.verma@example.com

```

2 rows in set (0.00 sec)

mysql> |
```

```
mysql> create table Departments (  
-> DeptID int primary key,  
-> DeptName varchar(50) not null  
-> );  
Query OK, 0 rows affected (0.03 sec)  
  
mysql> insert into Departments (Dept ID, DeptName) values (1, 'CSE');  
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right  
t syntax to use near 'ID, DeptName) values (1, 'CSE')' at line 1  
mysql> insert into Departments (DeptID, DeptName) values (1, 'CSE');  
Query OK, 1 row affected (0.01 sec)  
  
mysql> insert into Departments (DeptID, DeptName) values (2, 'ECE');  
Query OK, 1 row affected (0.01 sec)  
  
mysql> select * from Departments;  
+-----+-----+  
| DeptID | DeptName |  
+-----+-----+  
|      1 | CSE      |  
|      2 | ECE      |  
+-----+-----+  
2 rows in set (0.00 sec)  
  
mysql> |
```

```
mysql> create table Projects (
-> ProjectID int primary key,
-> ProjectName varchar(100) not null,
-> DeptID int,
-> FOREIGN KEY (DeptID) REFERENCES Departments(DeptID)
-> );
Query OK, 0 rows affected (0.05 sec)

mysql> insert into Projects (ProjectID, ProjectName, DeptID) values (201, 'Smart Expense Tracker', 1);
Query OK, 1 row affected (0.02 sec)

mysql> insert into Projects (ProjectID, ProjectName, DeptID) values (202, 'IoT Home Automation', 2);
Query OK, 1 row affected (0.01 sec)

mysql> select * from Projects;
```

ProjectID	ProjectName	DeptID
201	Smart Expense Tracker	1
202	IoT Home Automation	2

```
2 rows in set (0.00 sec)

mysql> |
```

```
mysql> create table Customers (
-> CustomerID int primary key,
-> CustomerName varchar(100) not null,
-> Email varchar(100) unique,
-> Phone varchar(15)
-> );
Query OK, 0 rows affected (0.06 sec)

mysql> insert into Customers (CustomerID, CustomerName, Email, Phone) values (301, 'Anil Kumar', 'anil.kumar@example.com', '9876543210');
Query OK, 1 row affected (0.01 sec)

mysql> insert into Customers (CustomerID, CustomerName, Email, Phone) values (302, 'Sneha Reddy', 'sneha.reddy@example.com', '7675937088');
Query OK, 1 row affected (0.01 sec)

mysql> select * from Customers;
```

CustomerID	CustomerName	Email	Phone
301	Anil Kumar	anil.kumar@example.com	9876543210
302	Sneha Reddy	sneha.reddy@example.com	7675937088

```

2 rows in set (0.00 sec)

mysql> |
```

```
mysql> create table Products (  
-> ProductID int primary key,  
-> ProductName varchar(100) not null,  
-> Price decimal(10,2) not null  
-> );
```

Query OK, 0 rows affected (0.03 sec)

```
mysql> insert into Products (ProductID, ProductName, Price) values (401, 'Laptop', 60000);  
Query OK, 1 row affected (0.01 sec)
```

```
mysql> insert into Products (ProductID, ProductName, Price) values(402, 'Smartphone', 25000);  
Query OK, 1 row affected (0.01 sec)
```

```
mysql> select * from Products;
```

ProductID	ProductName	Price
401	Laptop	60000.00
402	Smartphone	25000.00

2 rows in set (0.00 sec)

```
mysql> |
```

```
mysql> create table Orders (  
-> OrderID int primary key,  
-> CustomerID int not null,  
-> OrderDate DATE not null,  
-> FOREIGN KEY (CustomerID) REFERENCES Customers(CustomerID)  
-> );
```

Query OK, 0 rows affected (0.05 sec)

```
mysql> insert into Orders (OrderID, CustomerID, OrderDate) values (501, 301, '2026-08-01');  
Query OK, 1 row affected (0.01 sec)
```

```
mysql> insert into Orders (OrderID, CustomerID, OrderDate) values (502, 302, '2026-08-02');  
Query OK, 1 row affected (0.01 sec)
```

```
mysql> select * from Orders;
```

OrderID	CustomerID	OrderDate
501	301	2026-08-01
502	302	2026-08-02

2 rows in set (0.00 sec)

```
mysql> |
```

```

mysql> create table Order_Details (
  -> OrderDetailID int primary key,
  -> OrderID int not null,
  -> ProductID int not null,
  -> Quantity int not null,
  -> FOREIGN KEY (OrderID) REFERENCES Orders(OrderID),
  -> FOREIGN KEY (ProductID) REFERENCES Products(ProductID)
  -> );
Query OK, 0 rows affected (0.06 sec)

mysql> insert into Order_Details (OrderDetailID, OrderID, ProductID, Quantity) values (601, 401, 501, 1);
ERROR 1452 (23000): Cannot add or update a child row: a foreign key constraint fails ('practice2`.`order_details`, CONSTRAINT `order_details_ibfk_1` FOREIGN KEY (`OrderID`) REFERENCES `orders` (`OrderID`))
mysql> insert into Order_Details (OrderDetailID, OrderID, ProductID, Quantity) values (601, 501, 501, 1);
ERROR 1452 (23000): Cannot add or update a child row: a foreign key constraint fails ('practice2`.`order_details`, CONSTRAINT `order_details_ibfk_2` FOREIGN KEY (`ProductID`) REFERENCES `products` (`ProductID`))
mysql> insert into Order_Details (OrderDetailID, OrderID, ProductID, Quantity) values (601, 501, 401, 1);
Query OK, 1 row affected (0.01 sec)

mysql> insert into Order_Details (OrderDetailID, OrderID, ProductID, Quantity) values (602, 502, 402, 2);
Query OK, 1 row affected (0.01 sec)

mysql> select * from Order_Details;
+-----+-----+-----+-----+
| OrderDetailID | OrderID | ProductID | Quantity |
+-----+-----+-----+-----+
|          601 |       501 |         401 |         1 |
|          602 |       502 |         402 |         2 |
+-----+-----+-----+-----+
2 rows in set (0.00 sec)

mysql>

```



```
mysql> insert into Employees (EmpID, EmpName, DeptID, Salary, Email) values (104, 'Meena Gupta', 1, 48000, 'meena.gupta@example.com')  
;  
Query OK, 1 row affected (0.01 sec)
```

```
mysql> select * from Employees;
```

EmpID	EmpName	DeptID	Salary	Email
101	Rahul Sharma	1	55000.00	rahul.sharma@example.com
102	Priya Verma	2	45000.00	priya.verma@example.com
104	Meena Gupta	1	48000.00	meena.gupta@example.com

```
3 rows in set (0.00 sec)
```

```
mysql> |
```

```
UPDATE Employees
-> SET Salary = Salary * 1.10
-> WHERE DeptID = (SELECT DeptID FROM Departments WHERE DeptName = 'CSE');
Query OK, 2 rows affected (0.01 sec)
Rows matched: 2  Changed: 2  Warnings: 0
```

```
mysql> DELETE FROM Employees
-> WHERE Salary < 30000;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> select * from Employees;
```

EmpID	EmpName	DeptID	Salary	Email
101	Rahul Sharma	1	60500.00	rahul.sharma@example.com
102	Priya Verma	2	45000.00	priya.verma@example.com
104	Meena Gupta	1	52800.00	meena.gupta@example.com

3 rows in set (0.00 sec)

```
mysql> SELECT e.EmpName, d.DeptName
-> FROM Employees e
-> JOIN Departments d ON e.DeptID = d.DeptID;
```

EmpName	DeptName
Rahul Sharma	CSE
Priya Verma	ECE
Meena Gupta	CSE

3 rows in set (0.00 sec)

```
mysql> SELECT e.EmpName
-> FROM Employees e
-> LEFT JOIN Departments d ON e.DeptID = d.DeptID
-> WHERE d.DeptID IS NULL;
```

Empty set (0.00 sec)

```
mysql> SELECT d.DeptName, e.EmpName
-> FROM Departments d
-> LEFT JOIN Employees e ON d.DeptID = e.DeptID;
```

DeptName	EmpName
CSE	Meena Gupta
CSE	Rahul Sharma
ECE	Priya Verma

3 rows in set (0.00 sec)

```
mysql> SELECT e.EmpName AS Employee, m.EmpName AS Manager
-> FROM Employees e
-> JOIN Employees m ON e.ManagerID = m.EmpID;
```

```
mysql> SELECT COUNT(*) AS TotalEmployees FROM Employees;
```

TotalEmployees
3

```
1 row in set (0.01 sec)
```

```
mysql> SELECT d.DeptName, AVG(e.Salary) AS AvgSalary  
-> FROM Employees e  
-> JOIN Departments d ON e.DeptID = d.DeptID  
-> GROUP BY d.DeptName;
```

DeptName	AvgSalary
CSE	56650.000000
ECE	45000.000000

```
2 rows in set (0.00 sec)
```

```
mysql> SELECT d.DeptName  
-> FROM Employees e  
-> JOIN Departments d ON e.DeptID = d.DeptID  
-> GROUP BY d.DeptName  
-> HAVING AVG(e.Salary) > 60000;
```

```
Empty set (0.00 sec)
```

```
mysql> SELECT MAX(Salary) AS HighestSalary, MIN(Salary) AS LowestSalary  
-> FROM Employees;
```

HighestSalary	LowestSalary
60500.00	45000.00

```
1 row in set (0.00 sec)
```

```
mysql> |
```

```
mysql> use practice2;
```

```
Database changed
```

```
mysql> SELECT EmpName, Salary
```

```
-> FROM Employees
```

```
-> WHERE Salary > (SELECT AVG(Salary) FROM Employees);
```

```
+-----+-----+
| EmpName      | Salary  |
+-----+-----+
| Rahul Sharma | 60500.00 |
| Meena Gupta  | 52800.00 |
+-----+-----+
```

```
2 rows in set (0.01 sec)
```

```
mysql> SELECT EmpName
```

```
-> FROM Employees
```

```
-> WHERE DeptID = (SELECT DeptID FROM Employees WHERE EmpName = 'Rahul');
```

```
Empty set (0.00 sec)
```

```
mysql> SELECT EmpName, Salary
```

```
-> FROM Employees
```

```
-> WHERE Salary = (SELECT MAX(Salary) FROM Employees);
```

```
+-----+-----+
| EmpName      | Salary  |
+-----+-----+
| Rahul Sharma | 60500.00 |
+-----+-----+
```

```
1 row in set (0.00 sec)
```

```
mysql> SELECT DeptID, COUNT(*) AS NumEmployees
```

```
-> FROM Employees
```

```
-> GROUP BY DeptID
```

```
-> HAVING COUNT(*) > 5;
```

```
Empty set (0.00 sec)
```

```
mysql>
```

```
mysql> SELECT EmpName, Salary
-> FROM (
->     SELECT EmpName, Salary,
->           DENSE_RANK() OVER (ORDER BY Salary DESC) AS Sa
laryRank
->     FROM Employees
-> ) t
-> WHERE SalaryRank = 2;
```

EmpName	Salary
Meena Gupta	52800.00

1 row in set (0.00 sec)

```
mysql> SELECT e.EmpName, e.Salary, d.DeptName
-> FROM Employees e
-> JOIN Departments d ON e.DeptID = d.DeptID
-> WHERE e.Salary > (
->     SELECT AVG(Salary) FROM Employees WHERE DeptID = e.De
ptID
-> );
```

EmpName	Salary	DeptName
Rahul Sharma	60500.00	CSE

1 row in set (0.00 sec)

```
mysql> SELECT c.CustomerName
-> FROM Customers c
-> LEFT JOIN Orders o ON c.CustomerID = o.CustomerID
-> WHERE o.OrderID IS NULL;
```

Empty set (0.01 sec)

```
mysql> |
```

```
mysql> SELECT p.ProductName, SUM(od.Quantity) AS TotalSold
-> FROM Order_Details od
-> JOIN Products p ON od.ProductID = p.ProductID
-> GROUP BY p.ProductName
-> ORDER BY TotalSold DESC
-> LIMIT 1;
```

ProductName	TotalSold
Smartphone	2

1 row in set (0.00 sec)

```
mysql> SELECT d.DeptName
-> FROM Employees e
-> JOIN Departments d ON e.DeptID = d.DeptID
-> GROUP BY d.DeptName
-> ORDER BY COUNT(*) DESC
-> LIMIT 1;
```

DeptName
CSE

1 row in set (0.00 sec)

```
mysql> SELECT DATE_FORMAT(o.OrderDate, '%Y-%m') AS Month,
-> SUM(p.Price * od.Quantity) AS MonthlySales,
-> RANK() OVER (ORDER BY SUM(p.Price * od.Quantity) D
ESC) AS SalesRank
-> FROM Orders o
-> JOIN Order_Details od ON o.OrderID = od.OrderID
-> JOIN Products p ON od.ProductID = p.ProductID
-> GROUP BY Month;
```

```
mysql> SELECT c.CustomerName,
->          SUM(p.Price * od.Quantity) AS TotalPurchase,
->          RANK() OVER (ORDER BY SUM(p.Price * od.Quantity) D
DESC) AS RankPosition
-> FROM Customers c
-> JOIN Orders o ON c.CustomerID = o.CustomerID
-> JOIN Order_Details od ON o.OrderID = od.OrderID
-> JOIN Products p ON od.ProductID = p.ProductID
-> GROUP BY c.CustomerName;
```

CustomerName	TotalPurchase	RankPosition
Anil Kumar	60000.00	1
Sneha Reddy	50000.00	2

2 rows in set (0.00 sec)


```
mysql> SELECT Email, COUNT(*) AS CountDuplicate  
-> FROM Employees  
-> GROUP BY Email  
-> HAVING COUNT(*) > 1;  
Empty set (0.00 sec)
```

```
mysql> |
```